Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A PHASE 1, OPEN-LABEL, SINGLE-DOSE STUDY TO INVESTIGATE THE MASS BALANCE, METABOLISM AND EXCRETION OF [14C]-PF-07304814 IN HEALTHY PARTICIPANTS USING A 14C-MICROTRACER APPROACH **Short Title/Acronym:** 14C-PF-07304814 Mass Balance Study in Healthy Participants **Protocol Number:** C4611003 **NCT Number:** NCT05050682 **Sponsor Name:** Pfizer **Investigational Product:** PF-07304814 (Lufotrelvir) / [14C]-PF-07304814 **Phase of Development:** Phase 1 **Medical Monitor:** Pfizer Clinical Operations / Medical Monitor

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the mass balance, metabolism, and excretion profiles of [14C]-PF-07304814 (lufotrelvir) as a continuous intravenous infusion at a dose of 500 mg over 24 hours in healthy adult participants using a 14C-microtracer approach. **Secondary Objectives:** To evaluate the pharmacokinetics (PK) of PF-07304814 and its active moiety PF-00835231, and to assess the safety and tolerability of the drug.

2.2 Background & Rationale PF-07304814 (Lufotrelvir) is an experimental water-soluble phosphate ester prodrug that converts into PF-00835231, a potent inhibitor of the SARS-CoV-2 3CL protease (Mpro) designed for the treatment of COVID-19 in hospitalized patients. This study is medically and regulatory necessary to understand the human absorption, distribution, metabolism, and excretion (ADME) of the drug, specifically observing how it is cleared via endogenous metabolism and determining the structure of its metabolites.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 5 healthy participants **Number of Sites:** 1 Clinical Research Unit **Estimated Study Start:** 07-Oct-2021 **Estimated Primary Completion:** 10-Dec-2021

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy male participants. 2. Between the ages of 18 and 45 years. 3. Having a healthy body weight and willing to comply with all tests and inpatient confinement procedures.

4.2 Exclusion Criteria 1. Presence of any medical conditions or underlying disease. 2. Presence of active infections (including positive SARS-CoV-2 tests). 3. Unwillingness to comply with the intensive monitoring or sample collection procedures.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single 500 mg dose of PF-07304814 containing a [14C]-microtracer. **Route of Administration:** Continuous Intravenous (IV) Infusion. **Duration of Treatment:** 24-hour infusion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard care required for healthy volunteer clinical research management. **Prohibited:** Any concomitant medications, supplements, or dietary products that could interfere with metabolic pathways or pharmacokinetic assessments during the study period.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, Physical Exam, Baseline Labs
Baseline	Day 0	Admission to clinical research unit, baseline vitals
Intervention & Sampling	Days 1 to 10	24-hour IV Infusion, intensive blood sampling for PK, continuous urine and feces collection for mass balance and radiolabel recovery
End of Study	~2 Months	Final safety follow-up, Exit Interview

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Total recovery of administered [14C]-radioactivity in excreta (urine and feces) over the specified 10-day collection period; structural identification and quantification of metabolites in plasma and excreta. **Measurement Method:** Liquid chromatography with mass spectrometry (LC-MS) coupled with radiocarbon detection for metabolite profiling.

7.2 Secondary & Exploratory Endpoints * Maximum observed plasma concentration (C_{max}) of PF-07304814 and its active moiety PF-00835231. * Area under the plasma concentration-time curve (AUC) from time zero to the last quantifiable concentration. *

Incidence of treatment-emergent Adverse Events (AEs) and Serious Adverse Events (SAEs).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring by clinical staff during the inpatient confinement, physical exams, and lab assessments. **Serious Adverse Events (SAEs):** Evaluated and reported throughout the study period. No SAEs occurred in this trial. **Data Monitoring Committee (DMC):** Internal safety review by the Sponsor (Pfizer) to oversee healthy volunteer safety.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Concentration Set and Safety Analysis Set. **Sample Size Justification:** A sample size of $N=5$ participants is standard for Phase 1 ADME mass balance studies using a 14C-microtracer to adequately characterize mean PK and excretion parameters. **Handling of Missing Data:** Missing concentration values were not imputed; standard non-compartmental PK analysis methods were applied.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Intensive onsite monitoring and source data verification during the inpatient confinement phase. **Audit Trail:** Standard electronic Case Report Form (eCRF) locking and stringent quality/radiological audits by Pfizer QA.

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11. Study Identifiers & Governance

Primary ID: {{C4611003}} **Registry Number:** {{NCT05050682}} **Sponsor/Lead Organization:** {{Pfizer}} **Study Title:** {{A PHASE 1, OPEN-LABEL, SINGLE-DOSE STUDY TO INVESTIGATE THE MASS BALANCE, METABOLISM AND EXCRETION OF [14C]-PF-07304814 IN HEALTHY PARTICIPANTS USING A 14C-MICROTRACER APPROACH}} **Official Regulatory Title:** {{A PHASE 1, OPEN-LABEL, SINGLE-DOSE STUDY TO INVESTIGATE THE MASS BALANCE, METABOLISM AND EXCRETION OF [14C]-PF-07304814 IN HEALTHY PARTICIPANTS USING A 14C-MICROTRACER APPROACH}}

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12. Clinical Framework (Study Specific)

Primary Condition: {{COVID-19 (Evaluated in Healthy Participants)}} **Diagnosis Threshold:** {{Healthy Male Participants, Age 18-45}} **Medical Methodology:** {{Intravenous Administration of 3CL Protease Inhibitor Prodrug}} **Optimization Target:** {{Pharmacokinetic Profiling & Metabolite Clearance Tracking}}

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13. Study Procedures & Methodology

Management Pathway: {{Phase 1 Clinical Pharmacology / ADME Pathway}} **Education Protocol (HCP):** {{Protocol for handling 14C-microtracer and intensive PK sampling}} **Education Protocol (Patient):** {{Instructions for inpatient confinement and total excreta collection}} **Quality Audit Mechanism:** {{Clinical Research Unit (CRU) Source Data Verification}}

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14. Observation & Follow-Up Schedule

Total Duration: {{Approximately 2 Months}} **Onsite Visit Cadence:** {{Continuous confinement for dosing (Day 1) and excreta collection (up to Day 10)}} **Remote Monitoring Frequency:** {{N/A - Intensive Inpatient Monitoring}} **Checklist Review Interval:** {{Daily assessments during confinement}}

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] |** ___ **| *[DD-MMM-YYYY]* | | *Lead Statistician* | *[Name]* |** _____ | [DD-MMM-YYYY] |